

DUT Software Plane

Operating System

Native NVMe / NIC Drivers

native device contract

Control Plane

DUT-Visible Device Interface

Virtual BDFs, BARs, capabilities

exposed interface

FPGA Front-End

logical interface, event routing

control events

Host Software Back-End

virtualization, control mediation

Data Plane

DUT Memory
queues and payload buffers

payload

DMA Window / Transport

device DMA

Physical Device
NVMe SSD or 82580 NIC

device command

control / metadata

payload movement